

Intelligent Recruitment and Talent Analytics Platform

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**Database Systems Engineering and Distributed Backend Development
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Submitted by

2510080029 - Oddesar Sudeeptha

2510080037 - Lokrey Latika

2510080043 - Samineni Sai Siri

2510080061 - Muvva Kovidha

Under the guidance of

Dr.Laiphangbam Melinda



Department of Electronics and Communication Engineering

Koneru Lakshmaiah Education Foundation, Aziz Nagar

Aziz Nagar – 500075

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Abstract

The Intelligent Recruitment and Talent Analytics Platform is a web-based application developed to automate and streamline the recruitment process. Traditional recruitment systems often involve manual candidate screening, job posting management, and interview scheduling, which can be time-consuming and prone to errors. Modern recruitment platforms leverage web technologies and database systems to improve efficiency and accuracy in hiring processes [1].

The proposed system provides a centralized platform where candidates can register, apply for jobs, and track recruitment activities, while recruiters can manage job postings, review candidate applications, schedule interviews, and analyze recruitment performance. The platform incorporates a Talent Matching module that compares candidate skills with job requirements to support effective shortlisting and hiring decisions [2].

The application is implemented using HTML, CSS, Java Servlets, Apache Tomcat, JDBC, and PostgreSQL [3][4][5]. Recruitment data is stored and managed efficiently through a relational database system, while the analytics dashboard provides insights into recruitment performance and interview outcomes. The system improves recruitment efficiency, reduces manual effort, and supports data-driven hiring decisions.

List of Figures

Figure No.	Figure Title
Figure 2.1	Workflow of Intelligent Recruitment and Talent Analytics Platform
Figure 2.2	System Architecture of Intelligent Recruitment and Talent Analytics Platform
Figure 3.1	Home Page of Intelligent Recruitment Platform
Figure 3.2	Candidate Registration Module
Figure 3.3	Recruiter Login Module
Figure 3.4	Job Management Module
Figure 3.5	View Jobs Module
Figure 3.6	Job Application Module
Figure 3.7	Interview Scheduling Module
Figure 3.8	Talent Matching Module
Figure 3.9	Recruitment Analytics Dashboard
Figure 4.1	Candidate Table in PostgreSQL Database
Figure 4.2	Job Table in PostgreSQL Database
Figure 4.3	Application Table in PostgreSQL Database
Figure 4.4	Interview Table in PostgreSQL Database
Figure 4.5	Recruiter Table in PostgreSQL Database
Figure 5.1	Recruitment Analytics Output

List of Tables

Table No.	Table Title
Table 1.1	Project Modules
Table 3.1	Software Requirements
Table 3.2	Hardware Requirements
Table 4.1	Candidate Table Structure
Table 4.2	Job Table Structure
Table 4.3	Application Table Structure
Table 4.4	Interview Table Structure
Table 4.5	Recruiter Table Structure
Table 5.1	Testing Results

Table of Contents

Section	Page Numbers
ABSTRACT	2
LIST OF FIGURES	3
LIST OF TABLES	4
TABLE OF CONTENTS	5-6
CHAPTER 1: INTRODUCTION	7
1.1 Objectives	8
1.2 Project Modules	8
CHAPTER 2: METHODOLOGY	9
2.1 System Workflow	9
2.2 System Architecture	10
2.3 Candidate Management Module	11
2.4 Job Management Module	11
2.5 Job Application Module	11
2.6 Talent Matching Module	11
2.7 Interview Scheduling Module	12
2.8 Analytics Dashboard Module	12
CHAPTER 3: IMPLEMENTATION	13
3.1 Software Requirements	13
3.2 Hardware Requirements	13
3.3 Home Page	14
3.4 Candidate Registration Module	14
3.5 Recruiter Login Module	15

3.6 Job Management Module	15
3.7 View Jobs Module	16
3.8 Job Application Module	16
3.9 Interview Scheduling Module	17
3.10 Talent Matching Module	18
3.11 Analytics Dashboard	18
CHAPTER 4: DATABASE DESIGN	19
4.1 Candidate Table	19
4.2 Job Table	20
4.3 Application Table	21
4.4 Interview Table	22
4.5 Recruiter Table	23
CHAPTER 5: RESULTS AND DISCUSSION	25
5.1 Results Obtained	25
5.2 Discussion	26
5.3 Advantages	27
5.4 Testing Results	27
CHAPTER 6: CONCLUSION AND FUTURE SCOPE	28
6.1 Future Scope	28
REFERENCES	29

INTELLIGENT RECRUITMENT AND TALENT ANALYTICS PLATFORM

CHAPTER 1: INTRODUCTION

Recruitment is a critical organizational activity that directly impacts workforce quality and organizational performance. As the number of job applicants increases, organizations require efficient systems to manage candidate information, job postings, interview schedules, and hiring decisions. Traditional recruitment approaches often rely on spreadsheets, email communication, and manual record keeping, resulting in inefficiencies and increased administrative effort [8].

The Intelligent Recruitment and Talent Analytics Platform addresses these challenges by providing an integrated recruitment management system. The platform allows candidates to create profiles, recruiters to manage vacancies, and organizations to monitor recruitment activities through analytical reports. The integration of web technologies and database management systems provides a reliable and scalable recruitment solution [7].

The project also incorporates talent analytics concepts, which enable organizations to evaluate candidate-job compatibility through skill-based matching. This improves candidate selection and helps recruiters identify qualified applicants more efficiently [2].

1.1 Objectives

The major objectives of the project are:

1. To automate candidate registration and profile management.
2. To provide job posting and recruitment management facilities.
3. To enable candidates to apply for available jobs online.
4. To automate interview scheduling and tracking.
5. To implement skill-based talent matching.
6. To generate recruitment analytics and performance reports.
7. To maintain recruitment information securely using PostgreSQL.
8. To provide a user-friendly web interface for candidates and recruiters.

1.2 Project Modules

Module	Description
Candidate Management	Candidate Registration and Profile Management
Job Management	Create and Manage Job Openings
Job Application	Apply for Available Jobs
Interview Scheduling	Schedule Candidate Interviews
Talent Matching	Match Candidate Skills with Job Requirements
Analytics Dashboard	View Recruitment Statistics
Authentication	Recruiter Login System

Table 1.1: Project Modules

CHAPTER 2: METHODOLOGY

The Intelligent Recruitment and Talent Analytics Platform follows a modular architecture that separates different recruitment functions into independent modules. The system uses Java Servlets for backend processing, PostgreSQL for database management, and Apache Tomcat as the deployment server [3][4][5].

2.1 System Workflow



Figure 2.1: Workflow of Intelligent Recruitment and Talent Analytics Platform

2.2 System Architecture

The system follows a three-tier architecture consisting of presentation, application, and database layers. The frontend is developed using HTML and CSS. Java Servlets process user requests and implement business logic. PostgreSQL stores candidate, job, application, and interview data. The Analytics Dashboard retrieves information from the database and displays recruitment statistics.

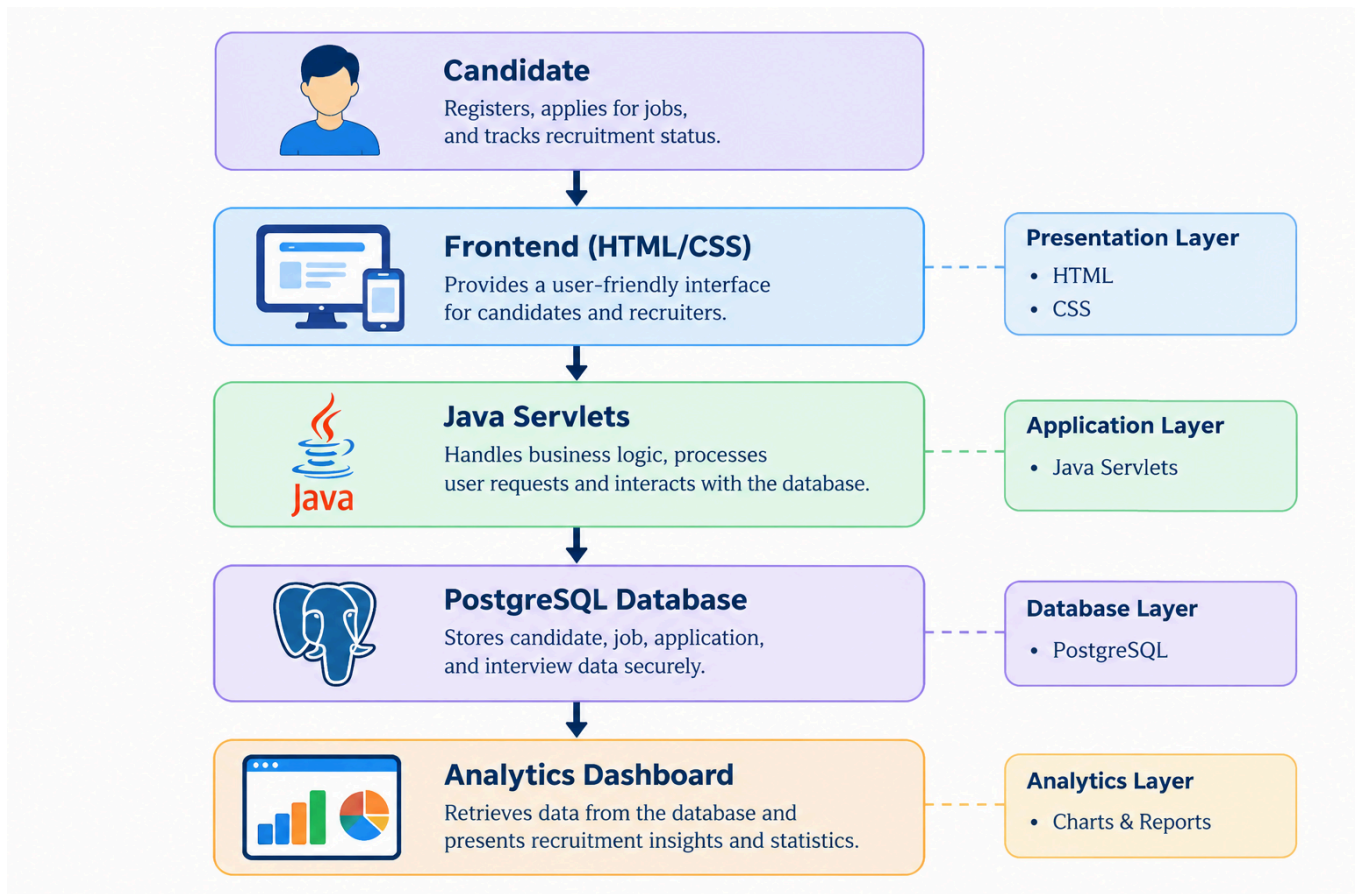


Figure 2.2: System Architecture of Intelligent Recruitment and Talent Analytics Platform

2.3 Candidate Management Module

The Candidate Management module allows job seekers to register their profiles by entering personal information, contact details, educational qualifications, and technical skills. The information is validated and stored in the PostgreSQL database using JDBC connectivity [4]. This module serves as the foundation for subsequent recruitment activities.

2.4 Job Management Module

Recruiters can create and manage job openings by specifying job titles, descriptions, required qualifications, experience levels, and skill requirements. This module helps organizations maintain an organized repository of available opportunities and simplifies vacancy management [1].

2.5 Job Application Module

Candidates can browse available job opportunities and submit applications directly through the platform. Application details are stored in the database and linked to corresponding job postings, enabling recruiters to review candidate submissions efficiently.

2.6 Talent Matching Module

Talent matching is a key feature of the platform. The system compares candidate skills against job requirements and generates a matching score. This assists recruiters in identifying candidates who best fit a particular position and supports data-driven recruitment decisions [2].

2.7 Interview Scheduling Module

The Interview Scheduling module allows recruiters to schedule candidate interviews, assign interview dates and times, and update interview status. Interview records are stored in the database, enabling recruiters to track recruitment progress effectively [10].

2.8 Analytics Dashboard Module

The Analytics Dashboard provides recruitment-related statistics such as total candidates, total jobs, total interviews, completed interviews, and cancelled interviews. These analytical insights support performance monitoring and strategic recruitment planning [2].

CHAPTER 3: IMPLEMENTATION

3.1 Software Requirements

S.No	Software	Purpose
1	Java Development Kit (JDK)	Java Application Development
2	Eclipse IDE	Coding and Project Development
3	Apache Tomcat Server	Deployment and Execution of Web Application
4	PostgreSQL	Database Management System
5	pgAdmin	Database Administration Tool
6	HTML	Frontend Structure Design
7	CSS	User Interface Styling
8	JDBC Driver	Database Connectivity

Table 3.1: Software Requirements

3.2 Hardware Requirements

S.No	Component	Specification
1	Processor	Intel Core i5 or Higher
2	RAM	Minimum 8 GB
3	Storage	256 GB or Higher
4	Operating System	Windows 11

Table 3.2: Hardware Requirements

3.3 Home Page

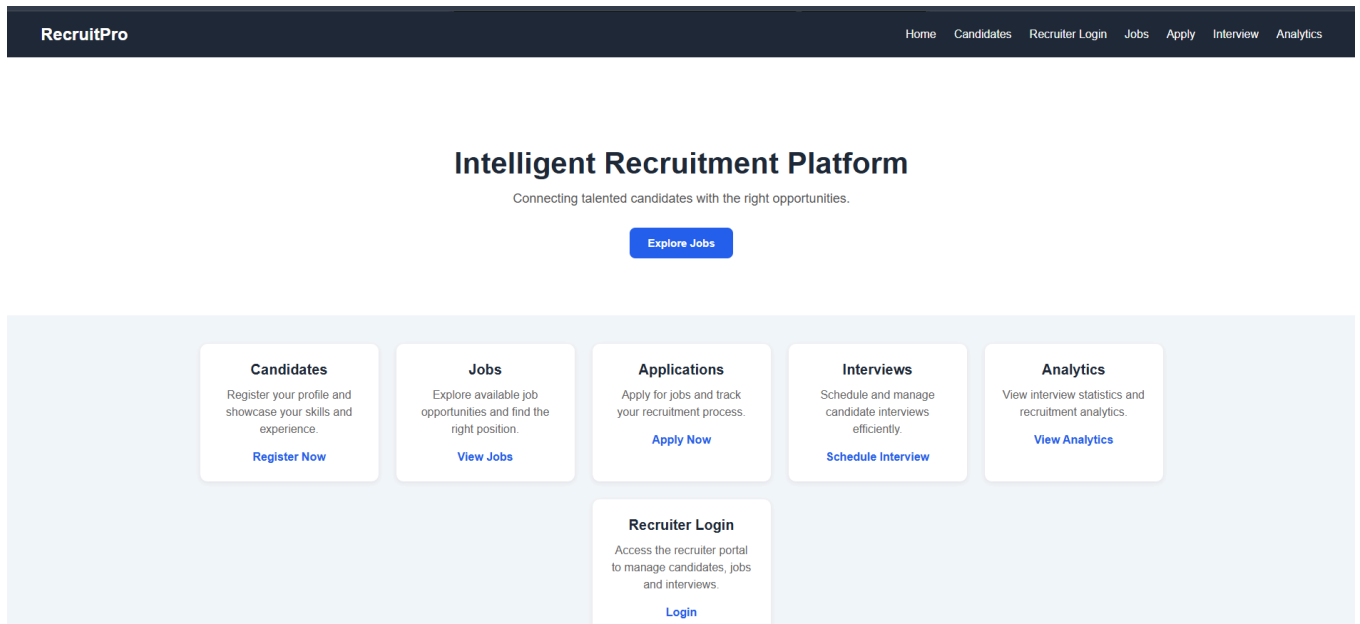


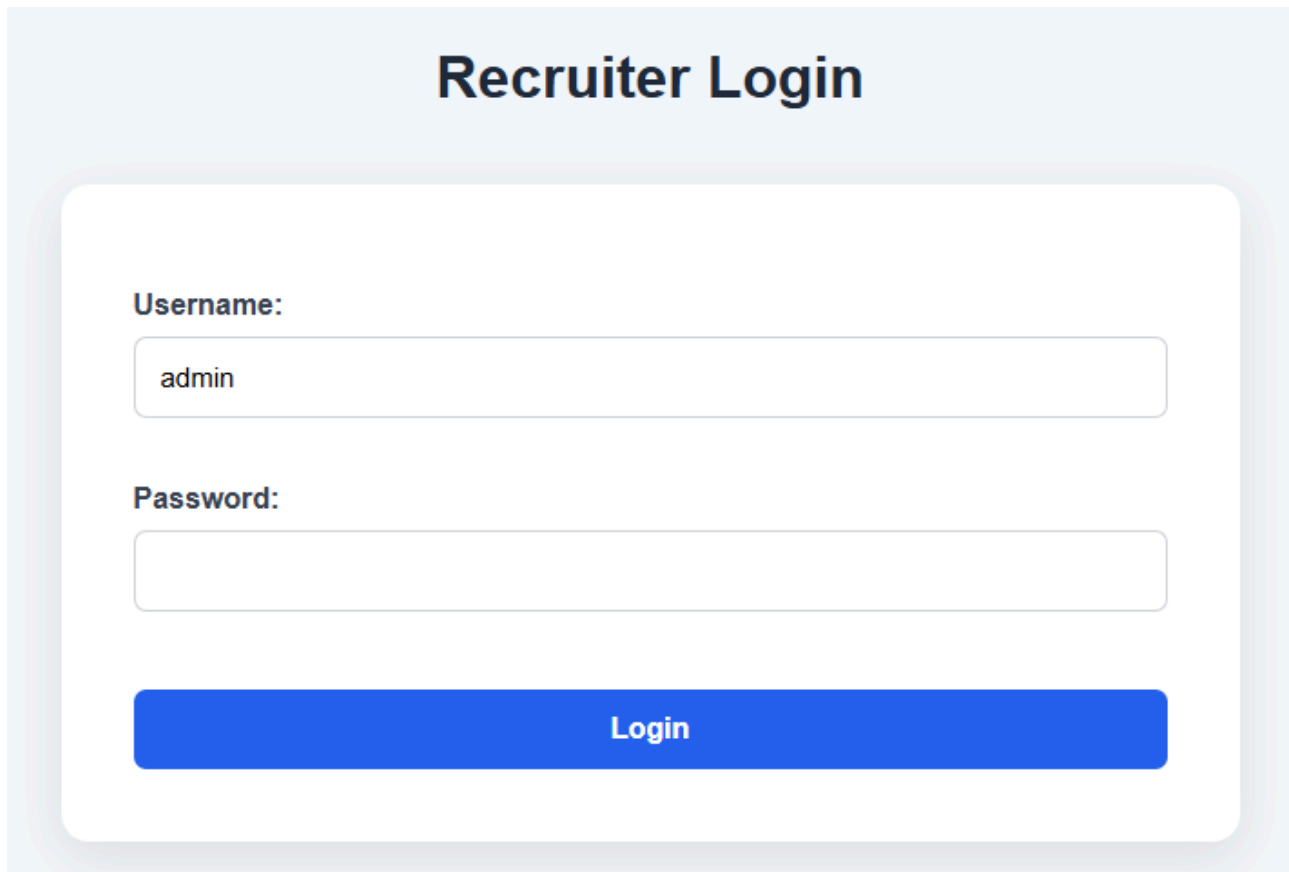
Figure 3.1: Home Page of Intelligent Recruitment Platform

3.4 Candidate Registration Module

The image displays the 'Candidate Registration' form. It has a clean, modern design with a light gray border. The title 'Candidate Registration' is centered at the top in a large, bold, dark blue font. Below the title are four labeled input fields: 'Name:', 'Email:', 'Phone:', and 'Skills:'. Each label is in a bold, dark blue font, and the corresponding input field is a simple white rectangle with a thin gray border. At the bottom of the form is a prominent blue button with the text 'Register Candidate' in white, bold font.

Figure 3.2: Candidate Registration Module

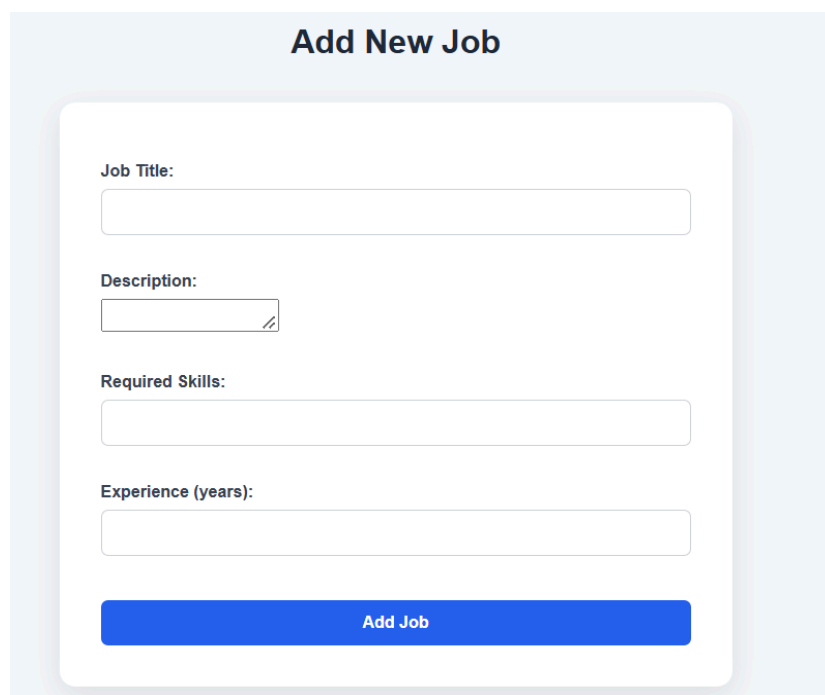
3.5 Recruiter Login Module



The Recruiter Login Module is a web form titled "Recruiter Login". It features a white rounded rectangle centered on a light blue background. Inside the rectangle, there are two input fields: "Username:" with the text "admin" and "Password:". Below these fields is a prominent blue button labeled "Login".

Figure 3.3: Recruiter Login Module

3.6 Job Management Module



The Add New Job form is titled "Add New Job" and is presented as a white rounded rectangle on a light blue background. It contains four input fields: "Job Title:", "Description:" (which includes a small icon for text formatting), "Required Skills:", and "Experience (years):". At the bottom of the form is a blue button labeled "Add Job".

Figure 3.4: Job Management Module

3.7 View Jobs Module

The screenshot shows a web application titled "Recruitment Platform" with a dark blue header. The header contains navigation links: Home, Jobs, Candidate, Interview, and Analytics. The main content area is titled "Available Jobs" and displays two job listings in white cards with blue borders. The first card is for a "Java Developer" position (Job ID: 1) with a description of "Backend development", required skills of "Java,SQL,DSA", and 2 years of experience. The second card is for a "Frontend Developer" position (Job ID: 2) with a description of "frontend development", required skills of "html,css,js", and 2 years of experience. Both cards feature a blue "Apply for this Job" button.

Recruitment Platform

Home Jobs Candidate Interview Analytics

Available Jobs

Java Developer
Job ID: 1
Description: Backend development
Required Skills: Java,SQL,DSA
Experience: 2 years
[Apply for this Job](#)

Frontend Developer
Job ID: 2
Description: frontend development
Required Skills: html,css,js
Experience: 2 years
[Apply for this Job](#)

Figure 3.5: View Jobs Module

3.8 Job Application Module

The screenshot shows a web application titled "Apply for a Job" in a large, bold, blue font. Below the title is a white rounded rectangle containing two input fields. The first field is labeled "Candidate ID:" and the second field is labeled "Job ID:". Below these fields is a large blue button with the text "Apply Now" in white.

Apply for a Job

Candidate ID:

Job ID:

[Apply Now](#)

Figure 3.6: Job Application Module

3.9 Interview Scheduling Module

Schedule Interview

Application ID:

Interview Date:

dd-mm-yyyy

Interview Time:

--:--

Interview Mode:

Online

Interviewer:

Schedule Interview

Figure 3.7: Interview Scheduling Module

3.10 Talent Matching Module

Talent Matching

Select a candidate and job to automatically calculate the skill compatibility score.

Candidate ID:

Job ID:

Find Match

Figure 3.8: Talent Matching Module

3.11 Analytics Dashboard

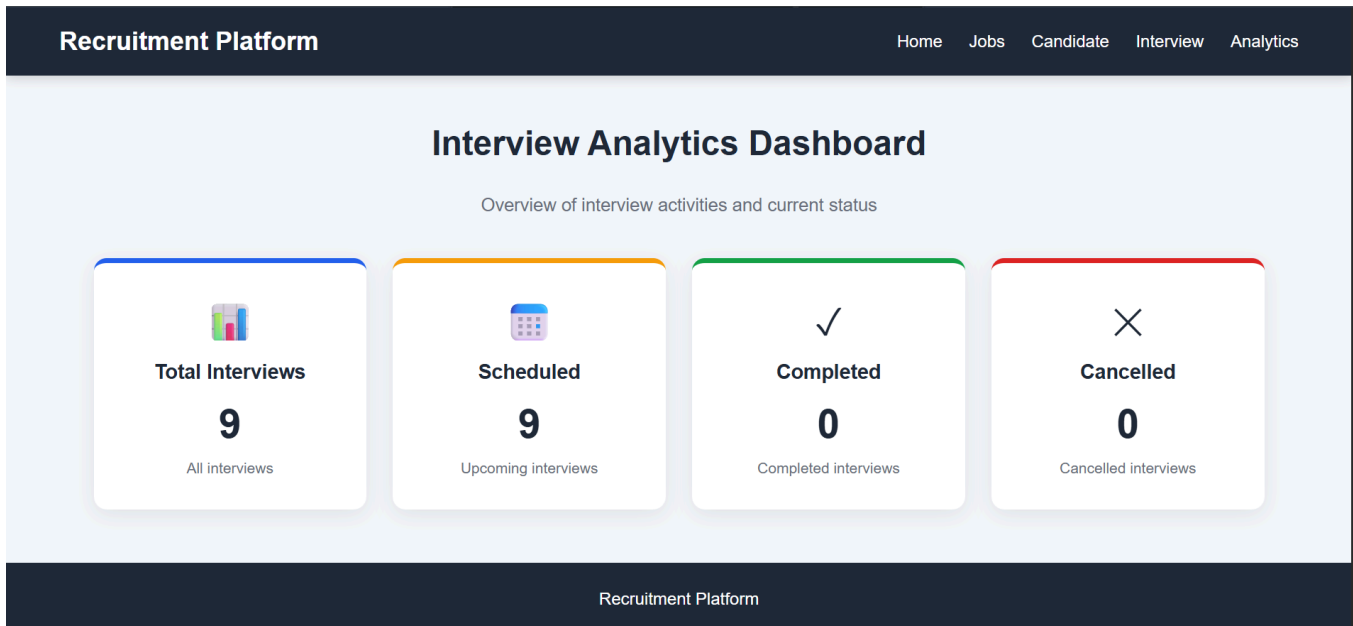


Figure 3.9: Recruitment Analytics Dashboard

Chapter 4: Database Design

The Intelligent Recruitment and Talent Analytics Platform uses PostgreSQL to store candidate, job, application, and interview information. The database is designed to maintain recruitment records efficiently and ensure data consistency [4].

4.1 Candidate Table

The Candidate table stores candidate information including personal details, contact information, educational qualifications, and skills. This information is used during candidate registration, job application, and talent matching processes. The table helps recruiters maintain a centralized repository of candidate profiles and facilitates efficient candidate management.

Field Name	Data Type	Constraints	Description
candidate_id	SERIAL	PRIMARY KEY	Unique Candidate ID
name	VARCHAR(100)	NOT NULL	Candidate Name
email	VARCHAR(100)	UNIQUE, NOT NULL	Candidate Email
phone	VARCHAR(15)	NOT NULL	Contact Number
skills	TEXT	NOT NULL	Candidate Skills

Table 4.1: Candidate Table Structure

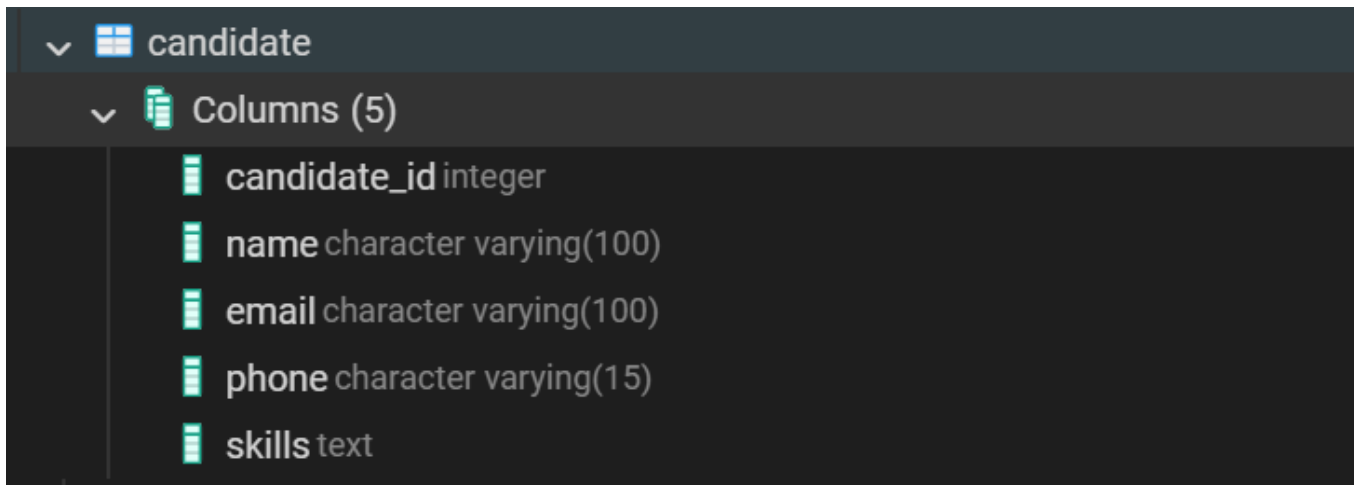


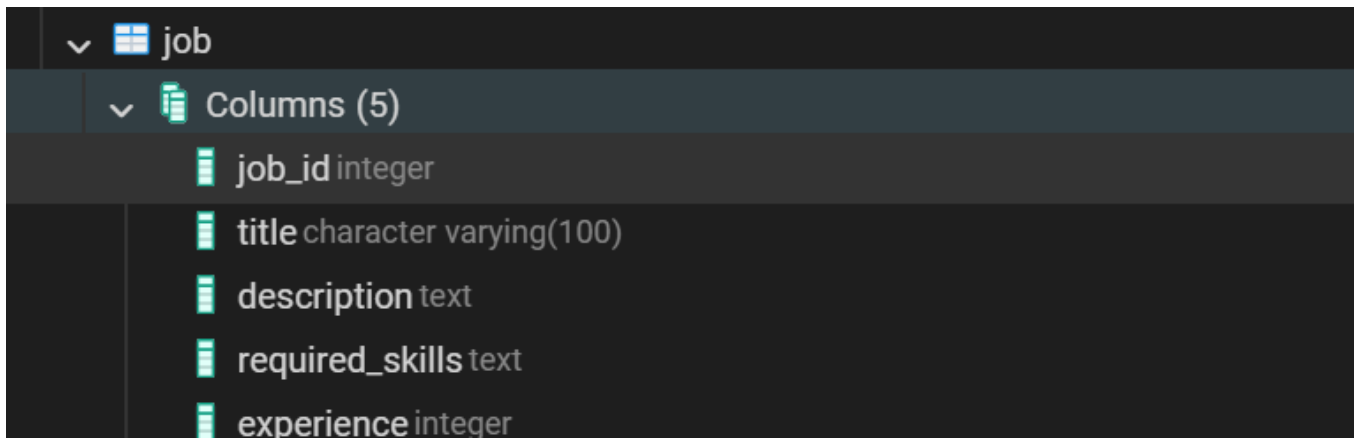
Figure 4.1: Candidate Table in PostgreSQL Database

4.2 Job Table

The Job table stores information related to job openings created by recruiters. It contains details such as job title, job description, required skills, and experience requirements. This table enables recruiters to manage available vacancies and helps candidates identify suitable job opportunities based on their qualifications and skills.

Field Name	Data Type	Constraints	Description
job_id	SERIAL	PRIMARY KEY	Unique Job ID
job_title	VARCHAR(100)	NOT NULL	Job Position
description	TEXT	NOT NULL	Job Description
skills_required	TEXT	NOT NULL	Required Skills
experience	VARCHAR(20)	NOT NULL	Experience Required

Table 4.2:: Job Table Structure



job	
Columns (5)	
job_id	integer
title	character varying(100)
description	text
required_skills	text
experience	integer

Figure 4.2: Job Table in PostgreSQL Database

4.3 Application Table

The Application Table maintains records of job applications submitted by candidates. It establishes a relationship between candidates and job postings by storing application details and status information. This table helps recruiters track applications and monitor candidate progress throughout the recruitment process.

Field Name	Data Type	Constraints	Description
application_id	SERIAL	PRIMARY KEY	Unique Application ID
candidate_id	INT	NOT NULL	Candidate ID
job_id	INT	NOT NULL	Job ID
application_date	DATE	NOT NULL	Date of Application
status	VARCHAR(20)	NOT NULL	Application Status

Table 4.3: Application Table Structure

application	
Columns (4)	
application_id	integer
candidate_id	integer
job_id	integer
status	character varying(30)

Figure 4.3: Application Table in PostgreSQL Database

4.4 Interview Table

The Interview table maintains records of candidate interviews throughout the recruitment process. It stores interview schedules, candidate details, job information, interview dates, and interview status. This table helps recruiters monitor recruitment progress and track interview outcomes effectively.

Field Name	Data Type	Constraints	Description
interview_id	SERIAL	PRIMARY KEY	Unique Interview ID
candidate_name	VARCHAR(100)	NOT NULL	Candidate Name
job_title	VARCHAR(100)	NOT NULL	Job Title
interview_date	DATE	NOT NULL	Interview Date
status	VARCHAR(20)	NOT NULL	Scheduled/Completed/ Cancelled

Table 4.4: Interview Table Structure

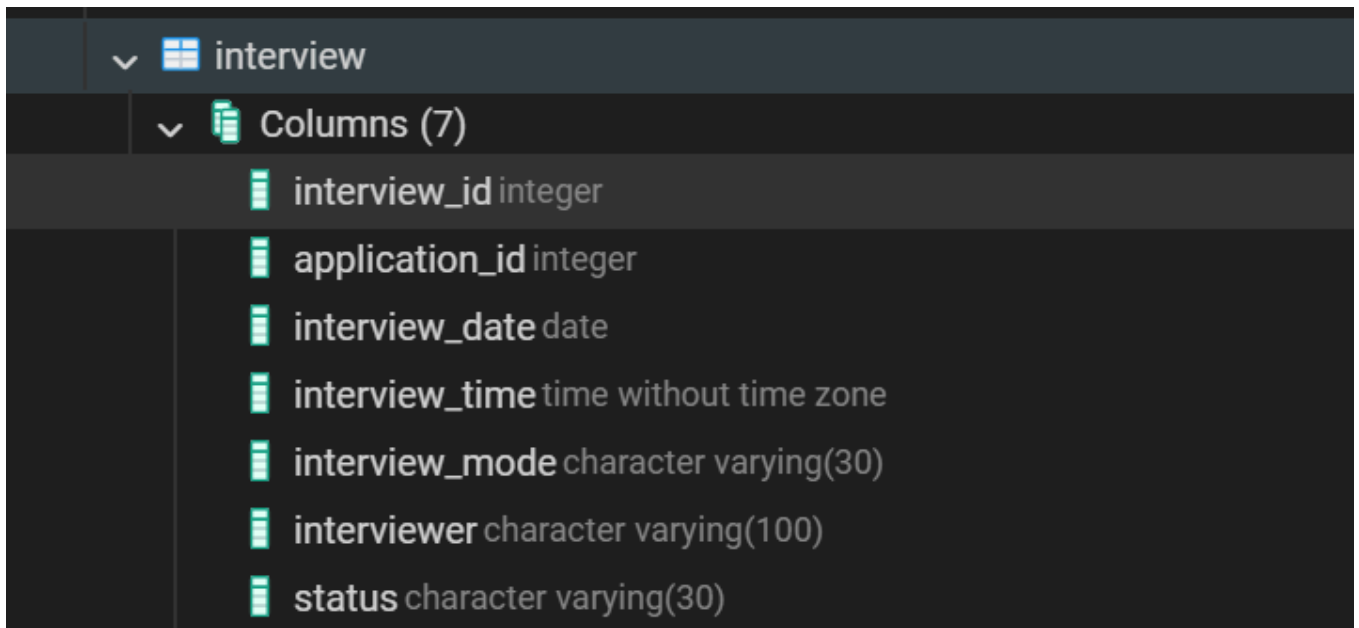


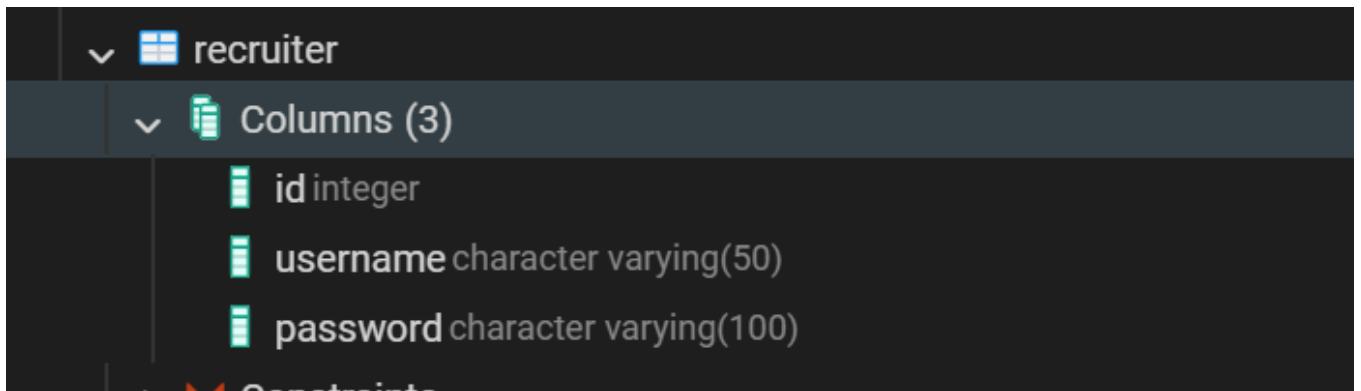
Figure 4.4: Interview Table in PostgreSQL Database

4.5 Recruiter Table

The Recruiter Table stores login credentials and account information for recruiters using the platform. It enables secure authentication and access control, ensuring that only authorized recruiters can manage job postings, candidate applications, interview schedules, and recruitment analytics.

Field Name	Data Type	Constraints	Description
recruiter_id	SERIAL	PRIMARY KEY	Unique Recruiter ID
username	VARCHAR(50)	UNIQUE, NOT NULL	Username
password	VARCHAR(100)	NOT NULL	Password

Table 4.5: Recruiter Table Structure



▼	recruiter
▼	Columns (3)
	id integer
	username character varying(50)
	password character varying(100)
	Constraints

Figure 4.5: Recruiter Table in PostgreSQL Database

CHAPTER 5: RESULTS AND DISCUSSION

The Intelligent Recruitment and Talent Analytics Platform was successfully developed and implemented using HTML, CSS, Java Servlets, JDBC, Apache Tomcat, and PostgreSQL. The system integrates candidate management, job management, talent matching, interview scheduling, and recruitment analytics into a unified platform. All modules were tested successfully and demonstrated proper interaction with the PostgreSQL database.

5.1 Results Obtained

The developed system achieved the following outcomes:

- Candidate profiles were successfully registered and stored in the PostgreSQL database.
- Recruiters were able to create, update, and manage job postings efficiently.
- Candidates could browse available job openings and submit applications online.
- Interview schedules were successfully created and maintained.
- Talent matching functionality assisted recruiters in identifying suitable candidates based on skills and job requirements.
- The Analytics Dashboard generated recruitment statistics including interview counts and recruitment progress.
- Data was retrieved and displayed correctly using JDBC connectivity between Java Servlets and PostgreSQL.

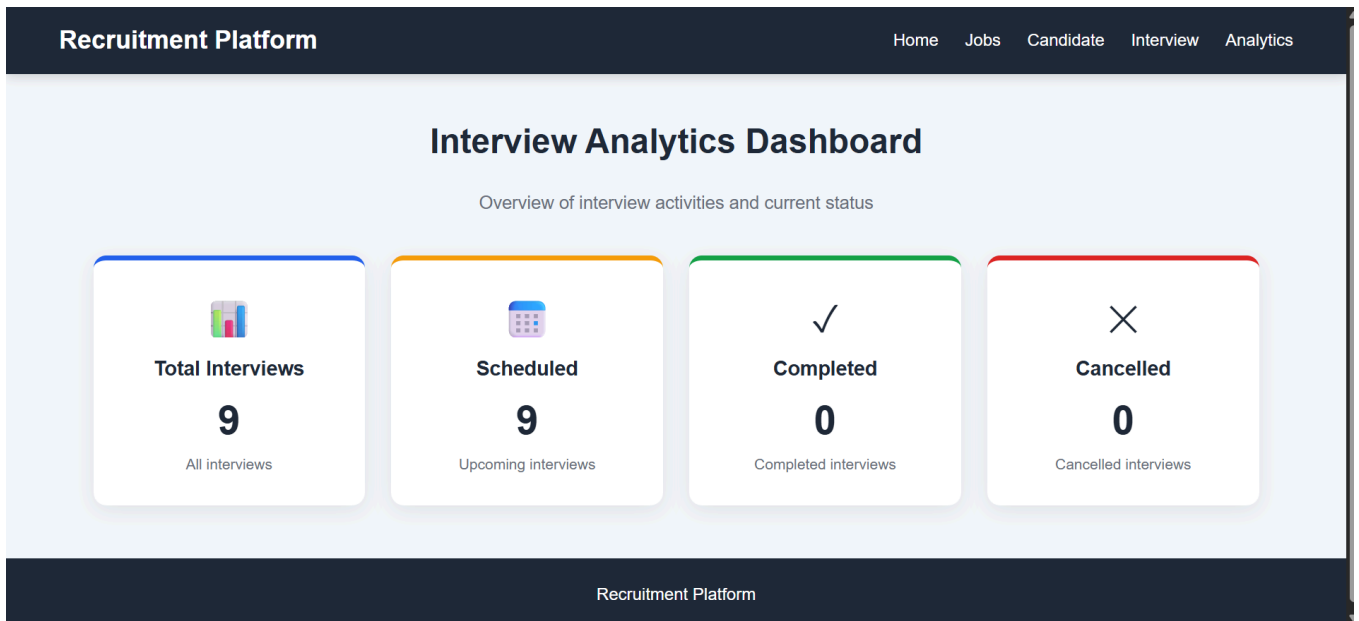


Figure 5.1: Recruitment Analytics Output

5.2 Discussion

The implementation demonstrates how modern web technologies can be utilized to automate recruitment activities effectively. The centralized database improves data organization and eliminates the need for manual record maintenance. The Candidate Management and Job Management modules streamline the recruitment workflow, while the Talent Matching module supports better candidate selection by comparing skills with job requirements [2].

The Interview Scheduling module improves recruitment tracking by maintaining interview records and statuses. The Analytics Dashboard provides valuable insights into recruitment activities, helping recruiters monitor hiring performance and make informed decisions. Overall, the system enhances efficiency, reduces manual effort, and supports structured recruitment management [7].

5.3 Advantages

The developed platform offers several advantages:

- Centralized recruitment management.
- Reduced paperwork and manual effort.
- Improved candidate-job matching.
- Efficient interview scheduling and tracking.
- Faster access to recruitment information.
- Better recruitment monitoring and reporting.
- User-friendly web-based interface.
- Secure storage of recruitment data using PostgreSQL.
- Improved decision-making through analytics and statistics.

5.4 Testing Results

Test Case	Expected Result	Status
Candidate Registration	Candidate Stored Successfully	Pass
Recruiter Login	Successful Authentication	Pass
Job Creation	Job Added Successfully	Pass
View Jobs	Jobs Displayed Correctly	Pass
Job Application	Application Submitted Successfully	Pass
Interview Scheduling	Interview Scheduled Successfully	Pass
Talent Matching	Matching Information Generated	Pass
Analytics Dashboard	Statistics Displayed Correctly	Pass

Table 5.1: Testing Results

Chapter 6: Conclusion And Future Scope

The Intelligent Recruitment and Talent Analytics Platform successfully integrates recruitment management, database systems, and web technologies into a unified application. The system automates candidate registration, job management, application processing, interview scheduling, and recruitment analytics. By reducing manual intervention and improving information accessibility, the platform enhances recruitment efficiency and decision-making [7][8].

The project demonstrates practical implementation of Java Servlets, PostgreSQL, JDBC connectivity, and web development technologies. It serves as an effective solution for modern recruitment management requirements and provides a foundation for future enhancements involving artificial intelligence and predictive analytics [2].

FUTURE SCOPE

- AI-Based Resume Screening
- Machine Learning Candidate Ranking
- Resume Parsing and Skill Extraction
- Email and SMS Notifications
- Video Interview Integration
- Advanced Recruitment Analytics
- Cloud-Based Deployment
- Integration with Professional Networking Platforms

These features would further improve recruitment efficiency and support intelligent hiring processes [2][10].

REFERENCE

- [1] S. Kumar and R. Sharma, “Online Recruitment Management System Using Web Technologies,” International Journal of Computer Applications, 2024.
- [2] P. Gupta and A. Verma, “Talent Analytics and Data-Driven Recruitment: A Modern Approach,” International Journal of Advanced Research in Computer Science, 2024.
- [3] Oracle Corporation, “Java Servlet Technology Documentation.” <https://docs.oracle.com>
- [4] PostgreSQL Global Development Group, “PostgreSQL Documentation.” <https://www.postgresql.org/docs>
- [5] Apache Software Foundation, “Apache Tomcat Documentation.” <https://tomcat.apache.org>
- [6] Eclipse Foundation, “Eclipse IDE Documentation.” <https://www.eclipse.org>
- [7] M. Fowler, *Patterns of Enterprise Application Architecture*. Addison-Wesley.
- [8] I. Sommerville, *Software Engineering*, 10th Edition, Pearson.
- [9] R. Pressman and B. Maxim, *Software Engineering: A Practitioner's Approach*, 9th Edition.
- [10] A. Rajput and S. Singh, “Web-Based Recruitment and Applicant Tracking System,” IJITEE, 2024.